

Tyler Buxton

(540) 354-1219 | tbux@vt.edu | tylerbuxton.com

EDUCATION

Virginia Tech

M.S. Thesis Computer Science & Applications, GPA: 4.0

May 2025

B.S. Computer Science, GPA: 3.88

May 2024

Societies: Lambda Chi Alpha, Phi Beta Kappa

PROJECTS

Git Interactive Visualization Learning Platform, M.S. Thesis, Virginia Tech

August 2023 – May 2025

- Developed a web-based platform to teach command line and Git through interactive visualizations, scaffolded exercises, and real-time feedback, supporting 5 instructors with unlimited customizable exercises
- Achieved a SUS score of 82.5 and an 8.8/10 expert recommendation score through a formal expert evaluation
- Pilot study demonstrated the platform's ability to differentiate skill levels, with experts completing exercises 3.6x faster (10 min vs. 35 min) than beginners

Empower, Personal Project

May 2025 – Present

- Built a multi-tenant scheduling platform with calendar-based weekly scheduling, time-off management, admin dashboards, and Stripe payment integration, designed to scale across unlimited tenant companies
- Developed full-stack monorepo using React, TypeScript, TanStack Router, React Query, and Tailwind CSS for the frontend
- Implemented 20+ type-safe API routes with Fastify and oRPC backed by AWS DynamoDB, secured with Clerk authentication and role-based authorization

AI Chatbot, Senior Capstone

January 2024 – May 2024

- Led 5 person team and collaborated with client to develop AI chatbot to comfort obesity and type 2 diabetes
- Integrated Llama and OpenAI LLMs to facilitate cue text generation for 100 users

RELEVANT EXPERIENCE

Full Stack Developer II, FedEx, Remote

June 2025 – Present

- Lead live maps implementation to automate weekly map installations for 50,000+ driver Android devices, eliminating ~52 hours of annual manual work and enabling international route support
- Delivered picture proof of delivery integration projected to save an estimated \$3M through reduced processing time and storage costs
- Automated multilingual localization pipeline to accurately translate app strings across English, French, Spanish, and additional languages for international markets
- Reduced Android codebase size by ~90% by introducing Lombok and HILT dependency injection, improving long-term maintainability

Graduate & Undergraduate Teaching Assistant, Blacksburg, VA

August 2022 – May 2025

- Worked as a TA for Software Design (CS 1114), Data Structures (CS 2114), and Comparative Languages (CS 3304)
- Improved leadership and communication skills by instructing labs of about 30 students

Software Engineer Intern, FedEx, Remote

June 2024 – August 2024

- Contributed to an Android mobile application used by over 50,000 drivers for pickup and delivery
- Gained experience with the agile software development process by attending daily standups, creating and completing stories, and participating in sprint planning, refinement, and retrospectives

Undergraduate Research Assistant, Blacksburg, VA - January 2023 – May 2024

- Analyzed 957 students' data using Python and Matplotlib to publish a paper in FIE 2024 conference
- Enhanced usability and added authentication workflow to React web application used to interactively teach command line and Git skills to Computer Science students
- Created backend used by instructors and students to manage classes and exercises

Software Engineer Intern, FedEx, Remote

June 2023 – August 2023

- Developed web application using Angular and TypeScript that communicated with enterprise REST APIs to reroute packages, reducing time spent rerouting packages by 50%
- Led biweekly demos for team members and presented final application to company vice presidents

SKILLS

Languages: Python, Java, TypeScript, JavaScript, HTML, CSS

Technologies: React, Android, Tailwind CSS, TanStack Router, React Query, Fastify, Zod, Vite, DynamoDB, Docker, Splunk, Claude